

**Practical Problem\_5:**

The following table gives the data on the rate of gross capital formation from 1965-66 to 1981-82 in Rs. Crores.

| Years   | Gross capital formation |
|---------|-------------------------|
| 1965-66 | 19.3                    |
| 1966-67 | 20.9                    |
| 1967-68 | 17.8                    |
| 1968-69 | 16.1                    |
| 1969-70 | 17.6                    |
| 1970-71 | 17.8                    |
| 1971-72 | 18.3                    |
| 1972-73 | 17.3                    |
| 1973-74 | 21.4                    |
| 1974-75 | 19.3                    |
| 1975-76 | 18.1                    |
| 1976-77 | 19.5                    |
| 1977-78 | 19.2                    |
| 1978-79 | 22.2                    |
| 1979-80 | 20.9                    |
| 1980-81 | 21.5                    |
| 1981-82 | 21.9                    |

- Plot the data on a graph paper and comment on the movement of the curve.
- Determine the trend values by Semi-average method.
- Determine the trend values by Moving-average method.
- Determine the trend values by Mathematical curve fitting method (least square method).
- Test the randomness of the data.

**Solution:**

a)

Comment: The curve obtained by plotting the given time series data shows the presence of (an) upward/downward/ no trend.

b) The series contains data for the 17 years, hence  $17=2k+1$ ,  $k=8$ . We take the first 8 years data to form the first half of the series and the last 8 years data to form the last half of the series. The years 1973-74 is not included in either of the half of the series. The average of the first half series has to be plotted against the midpoint of 1968-69 to 1969-70 and the last half series average has to be plotted against the midpoint of 1977-78 to 1978-79 and joining these two points by a straight line.

From the plot, we get the following trend values with the help of semi-average method.

| <b>Years</b> | <b>Trend values</b> |
|--------------|---------------------|
| 1965-66      |                     |
| 1966-67      |                     |
| 1967-68      |                     |
| 1968-69      |                     |
| 1969-70      |                     |
| 1970-71      |                     |
| 1971-72      |                     |
| 1972-73      |                     |
| 1973-74      |                     |
| 1974-75      |                     |
| 1975-76      |                     |
| 1976-77      |                     |
| 1977-78      |                     |
| 1978-79      |                     |
| 1979-80      |                     |
| 1980-81      |                     |
| 1981-82      |                     |

c) From the plot we see that there are 5 peaks and 5 troughs. The peaks intervals are 5,2,3,2 years respectively. The trough intervals are 4,3,2,2 years respectively. Hence, we see that 2 years common in the both cases. So, we consider 2 years moving average point.

| <b>Years</b> | <b>Gross capital formation</b> | <b>2 yearly moving average</b> | <b>Centered moving average (Trend)</b> |
|--------------|--------------------------------|--------------------------------|----------------------------------------|
| 1965-66      | 19.3                           | 20.10                          |                                        |
| 1966-67      | 20.9                           | 19.35                          | 19.73                                  |
| 1967-68      | 17.8                           | 16.95                          | 18.15                                  |
| 1968-69      | 16.1                           |                                |                                        |
| 1969-70      | 17.6                           |                                |                                        |
| 1970-71      | 17.8                           |                                |                                        |
| 1971-72      | 18.3                           |                                |                                        |
| 1972-73      | 17.3                           |                                |                                        |
| 1973-74      | 21.4                           |                                |                                        |
| 1974-75      | 19.3                           |                                |                                        |
| 1975-76      | 18.1                           |                                |                                        |
| 1976-77      | 19.5                           |                                |                                        |
| 1977-78      | 19.2                           |                                |                                        |
| 1978-79      | 22.2                           |                                |                                        |
| 1979-80      | 20.9                           |                                |                                        |
| 1980-81      | 21.5                           |                                |                                        |
| 1981-82      | 21.9                           |                                |                                        |

d) The trend line can be fitted to the given data by the method of least squares. Coding the years as 0,1,2,...,16. Now we consider the straight line  $Y = \alpha + \beta X + \epsilon$

where  $\hat{\alpha} = \text{?????}$  and  $\hat{\beta} = \text{?????}$

The fitted line is  $\hat{Y} = \hat{\alpha} + \hat{\beta}X$

| Years   | Coded Year | Observed values | Trend value ( $\hat{Y} = \hat{\alpha} + \hat{\beta}X$ ) |
|---------|------------|-----------------|---------------------------------------------------------|
| 1965-66 | 0          | 19.3            |                                                         |
| 1966-67 | 1          | 20.9            |                                                         |
| 1967-68 | 2          | 17.8            |                                                         |
| 1968-69 | 3          | 16.1            |                                                         |
| 1969-70 | 4          | 17.6            |                                                         |
| 1970-71 | 5          | 17.8            |                                                         |
| 1971-72 | 6          | 18.3            |                                                         |
| 1972-73 | 7          | 17.3            |                                                         |
| 1973-74 | 8          | 21.4            |                                                         |
| 1974-75 | 9          | 19.3            |                                                         |
| 1975-76 | 10         | 18.1            |                                                         |
| 1976-77 | 11         | 19.5            |                                                         |
| 1977-78 | 12         | 19.2            |                                                         |
| 1978-79 | 13         | 22.2            |                                                         |
| 1979-80 | 14         | 20.9            |                                                         |
| 1980-81 | 15         | 21.5            |                                                         |
| 1981-82 | 16         | 21.9            |                                                         |

e) **Null hypothesis:** The data was collected in a random manner.

**Alternative hypothesis:** The data was not collected in a random manner.

| Years   | Gross capital formation | Binary | Counting Runs |
|---------|-------------------------|--------|---------------|
| 1965-66 | 19.3                    |        |               |
| 1966-67 | 20.9                    |        |               |
| 1967-68 | 17.8                    |        |               |
| 1968-69 | 16.1                    |        |               |
| 1969-70 | 17.6                    |        |               |
| 1970-71 | 17.8                    |        |               |
| 1971-72 | 18.3                    |        |               |
| 1972-73 | 17.3                    |        |               |
| 1973-74 | 21.4                    |        |               |
| 1974-75 | 19.3                    |        |               |
| 1975-76 | 18.1                    |        |               |
| 1976-77 | 19.5                    |        |               |
| 1977-78 | 19.2                    |        |               |
| 1978-79 | 22.2                    |        |               |
| 1979-80 | 20.9                    |        |               |
| 1980-81 | 21.5                    |        |               |
| 1981-82 | 21.9                    |        |               |

Mean of **Gross capital formation (M)** =19.359

Under the null hypothesis of randomness, the **expected number of runs** is:

$$E(R) = 1 + \frac{2n_0n_1}{n_0 + n_1} =$$

$$Var(R) = \frac{2n_0n_1(2n_0n_1 - n_0 - n_1)}{(n_0 + n_1)^2 (n_0 + n_1 - 1)} =$$

Then compute the Z- statistics  $Z = \frac{R - E(R)}{\sqrt{Var(R)}} =$

**P-Value:**

**Comment:**